Name :- Juhi Shah

Subject :- (701) Application Development using Full Stack

Roll No :- Honours(12)

Semester :- 7

Division :- B

Assignment :- Practical Assignment 02

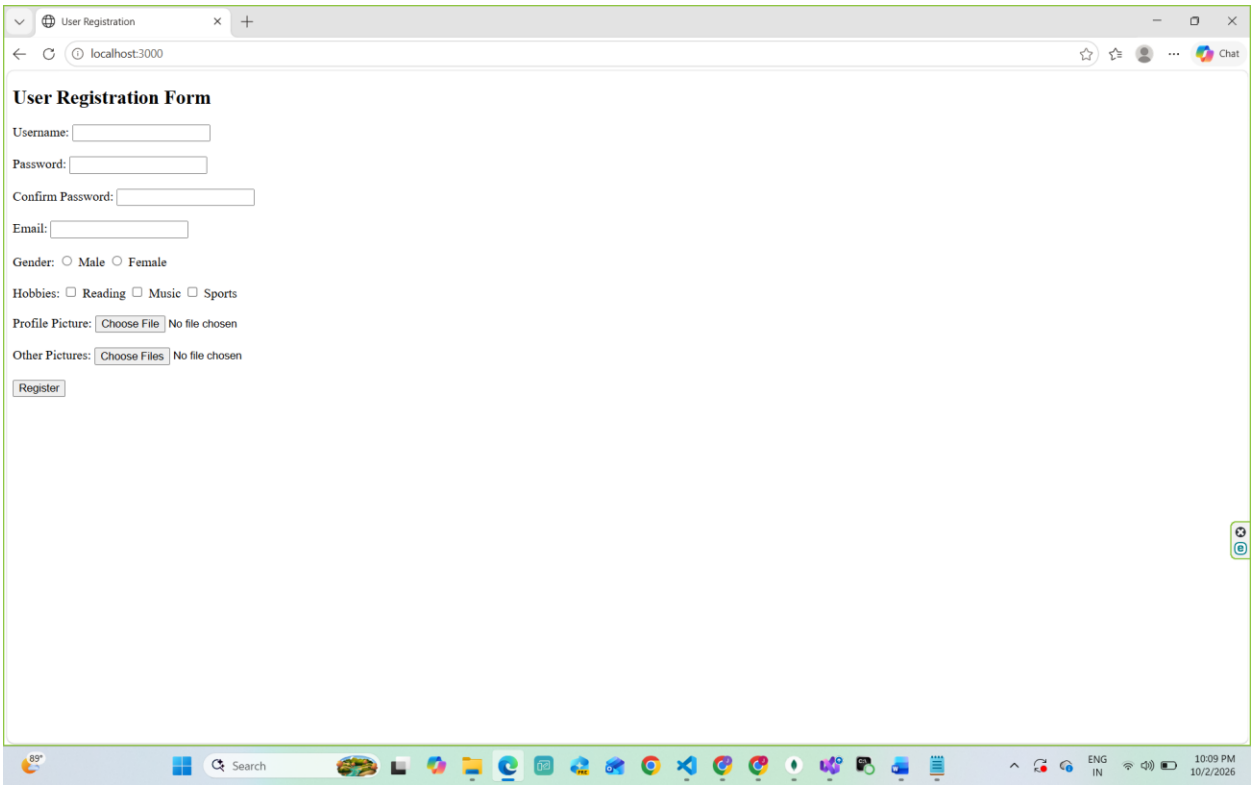
1) - Develop a user registration form (fields: username, password, confirm password, email, gender, hobbies, upload profile pic, upload other pics) (use text box, radio, checkboxes, file upload) using Express. Form should also contain file upload (single, multiple) with validations.

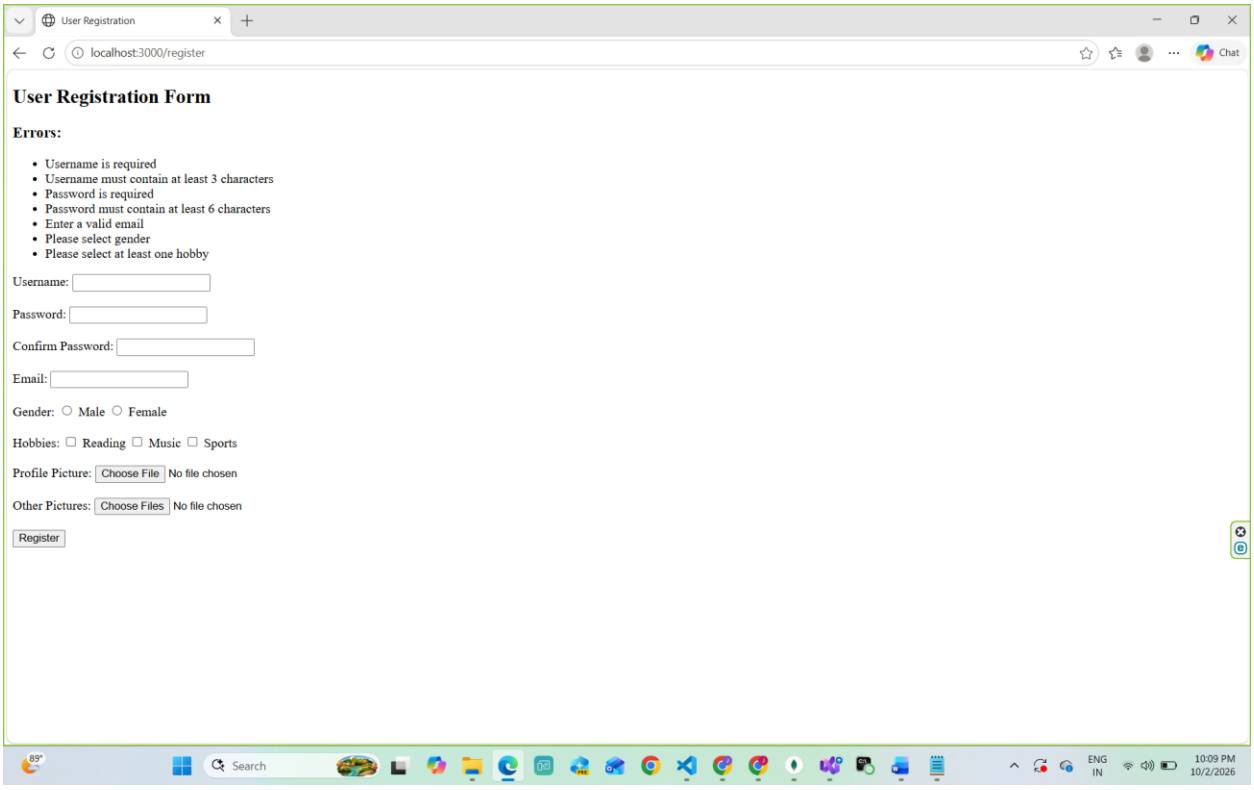
- In case of invalid data, display errors along with all previous field values.

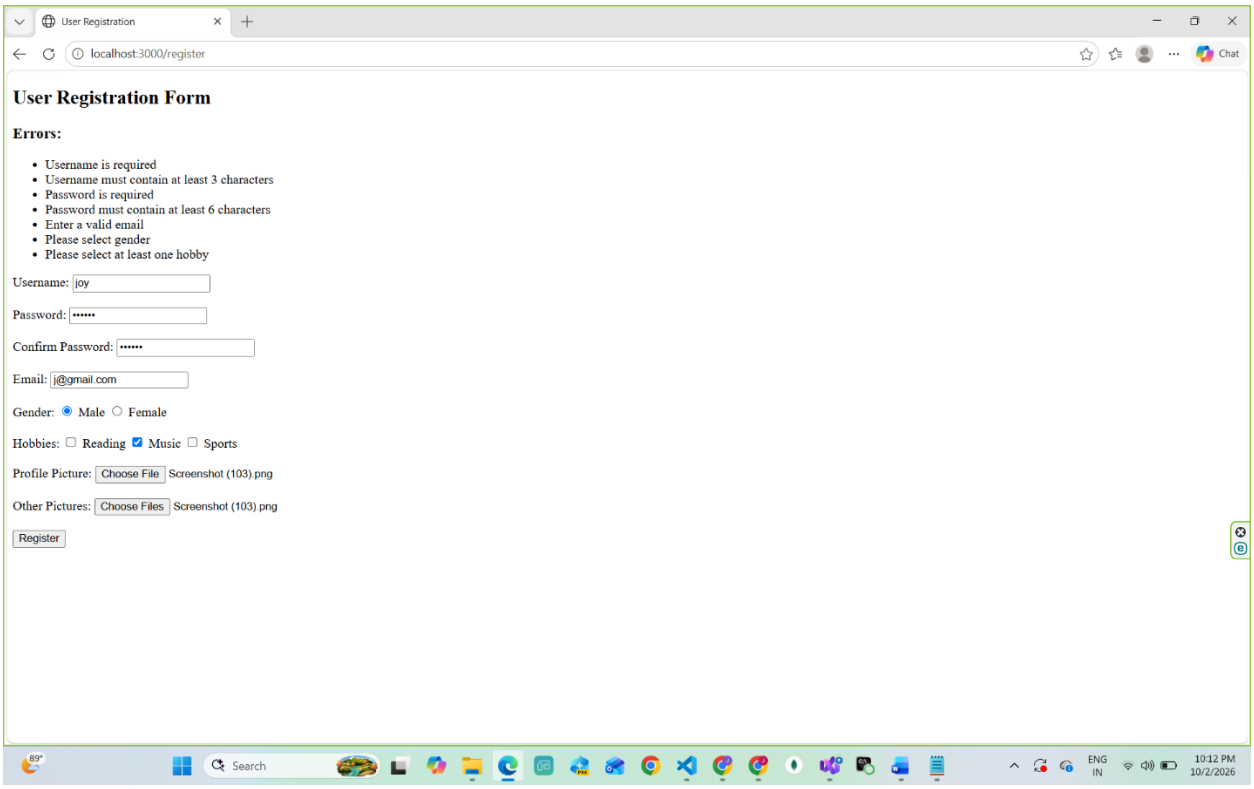
- In case of all valid fields, display all data and images in a well formatted tabular form.

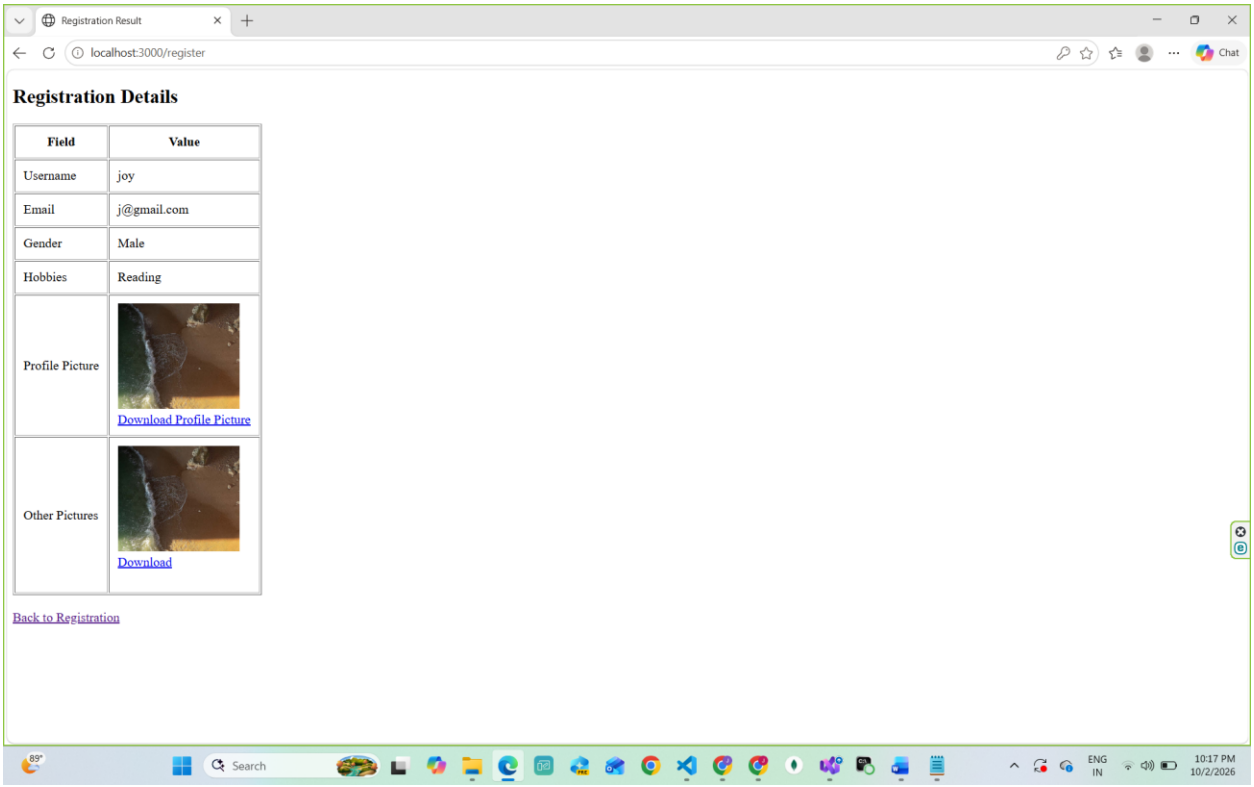
and allow the user to download that file using Express. Develop route for file download.

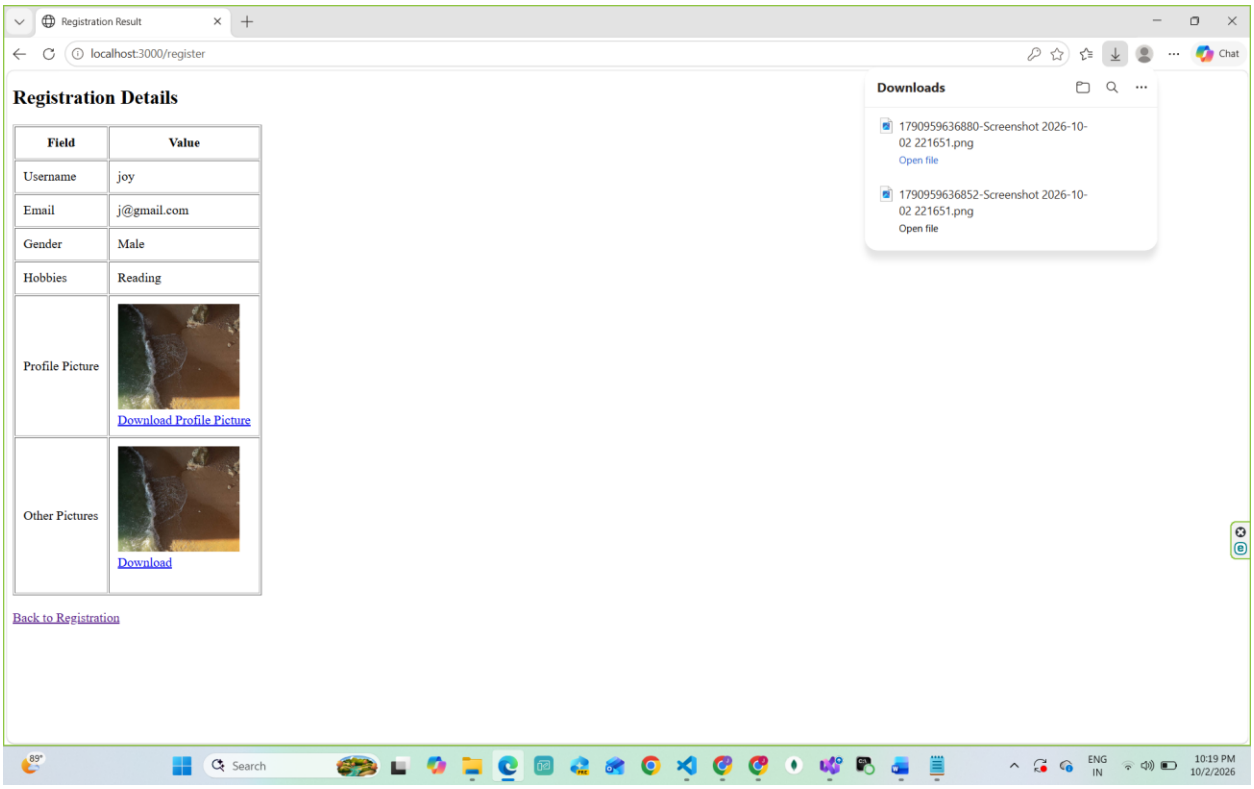
(Use express, ejs, express-validator)


User Registration
localhost:3000
Chat
User Registration Form
Username:
Password:
Confirm Password:
Email:
Gender: Male Female
Hobbies: Reading Music Sports
Profile Picture: Choose File No file chosen
Other Pictures: Choose Files No file chosen
Register
Search
ENG
IN
10:09 PM
10/2/2026

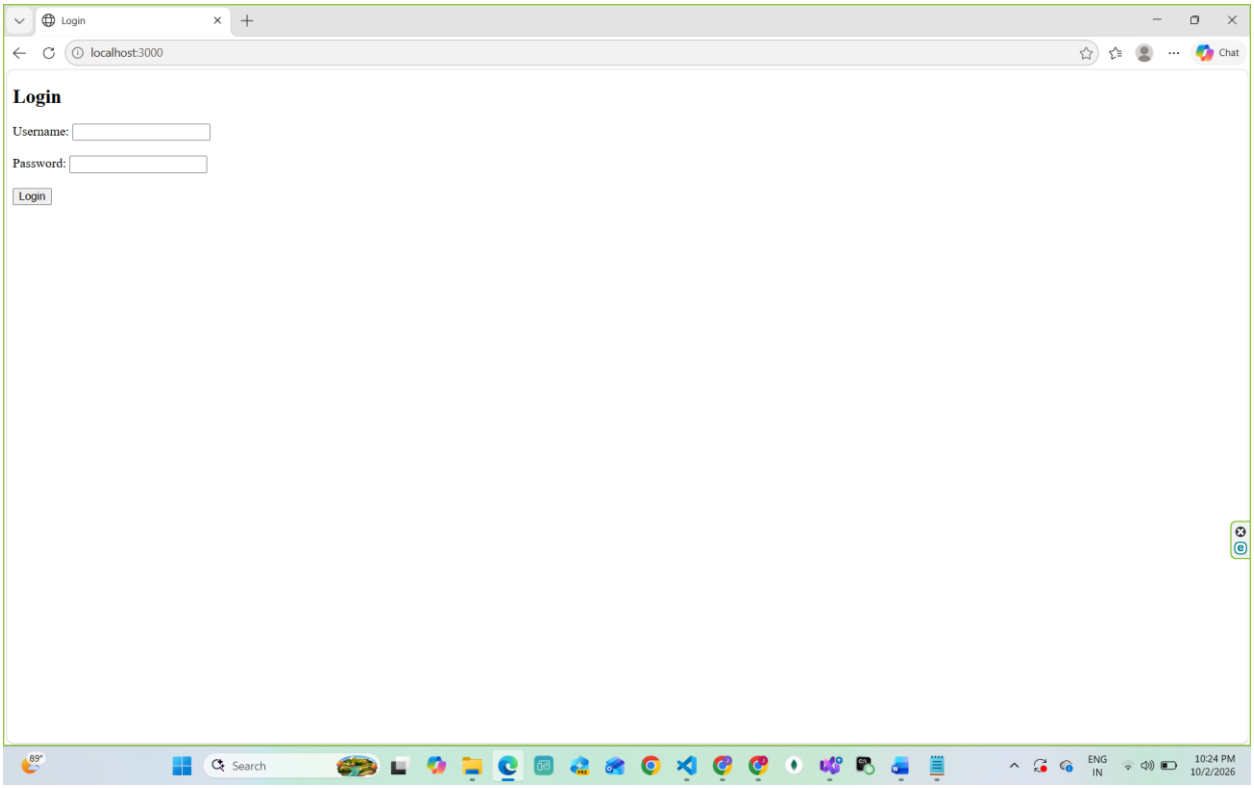

User Registration
localhost:3000/register
Chat
User Registration Form
Errors:
Username is required
Username must contain at least 3 characters
Password is required
Password must contain at least 6 characters
Enter a valid email
Please select gender
Please select at least one hobby
Username:
Password:
Confirm Password:
Email:
Gender: Male Female
Hobbies: Reading Music Sports
Profile Picture: Choose File No file chosen
Other Pictures: Choose Files No file chosen
Register
Search
ENG
IN
10:09 PM
10/2/2026

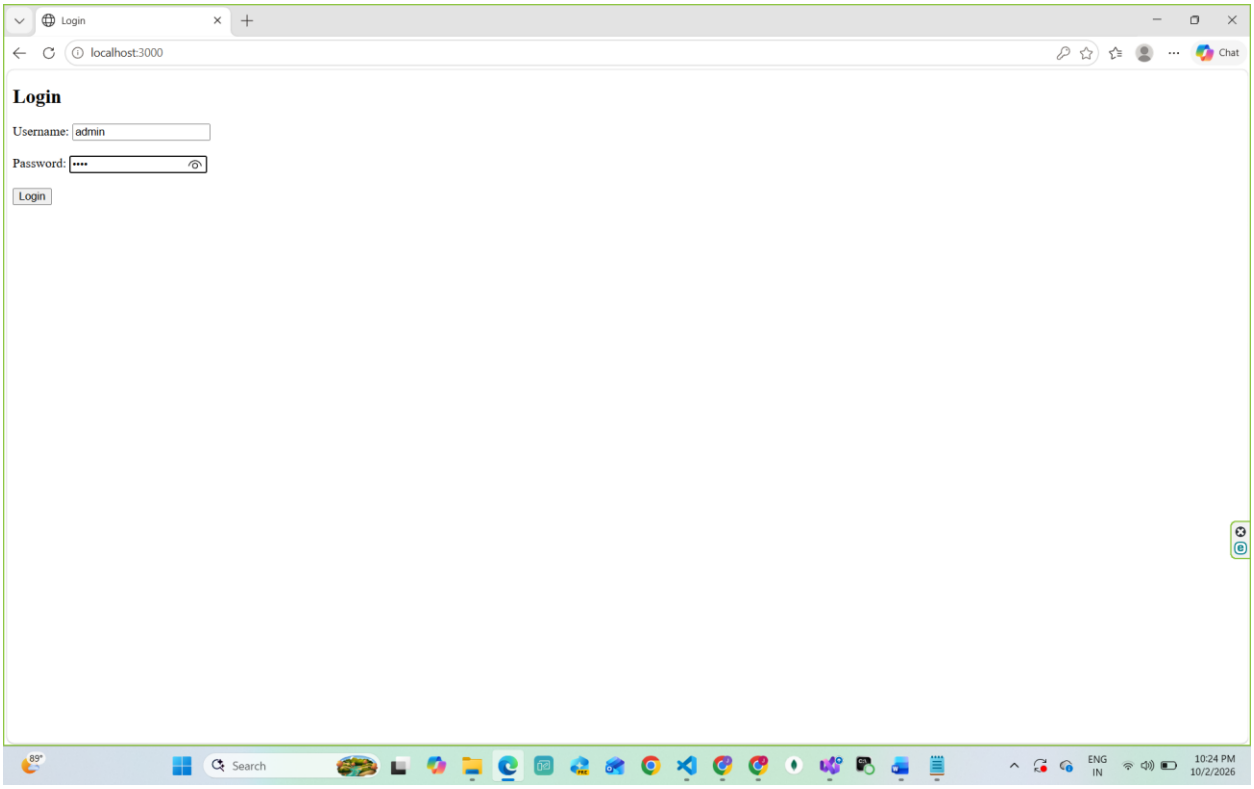

User Registration
localhost:3000/register
Chat
User Registration Form
Errors:
Username is required
Username must contain at least 3 characters
Password is required
Password must contain at least 6 characters
Enter a valid email
Please select gender
Please select at least one hobby
Username: joy
Password: ••••••
Confirm Password: ••••••
Email: j@gmail.com
Gender: Male Female
Hobbies: Reading Music Sports
Profile Picture: Choose File Screenshot (103).png
Other Pictures: Choose Files Screenshot (103).png
Register
Search
ENG IN
10:12 PM 10/2/2026

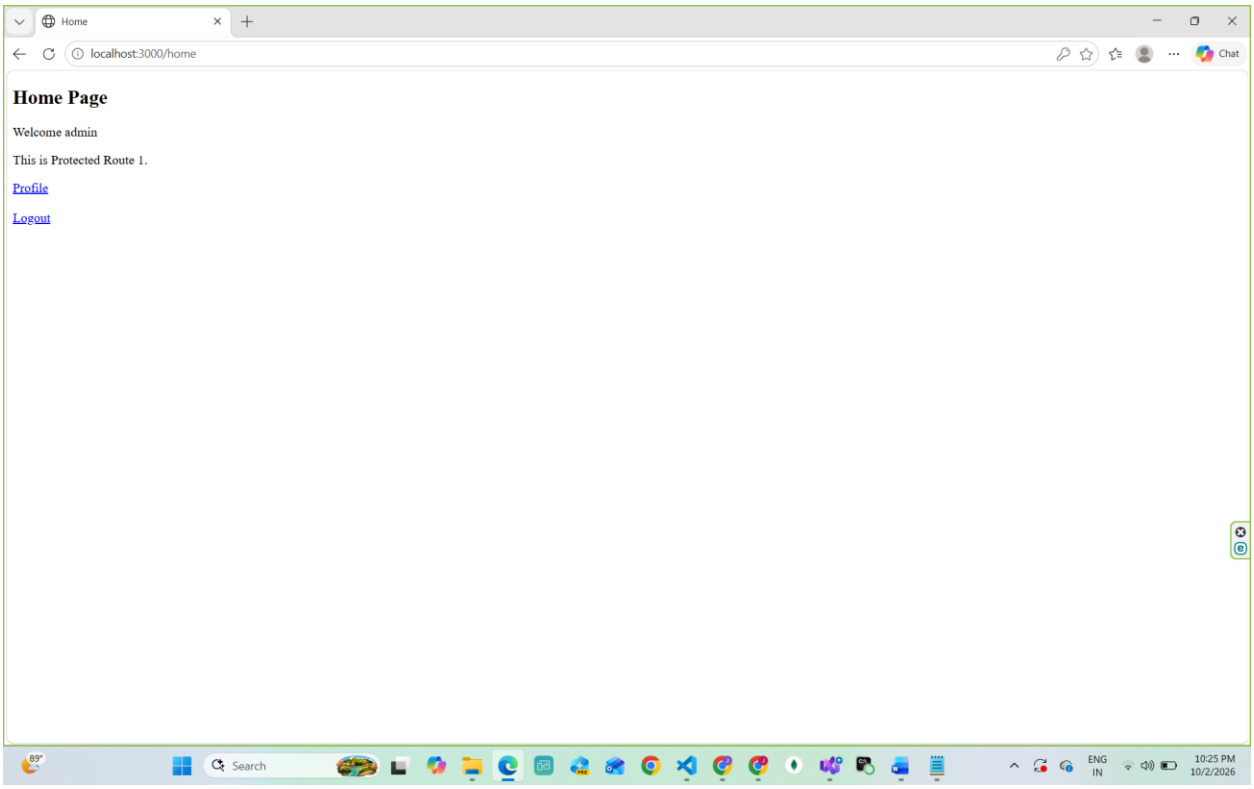

Registration Result
localhost:3000/register
Chat
Registration Details
Field
Value
Username
joy
Email
j@gmail.com
Gender
Male
Hobbies
Reading
Profile Picture
Download Profile Picture
Other Pictures
Download
Back to Registration
Search
ENG IN
10:17 PM 10/2/2026

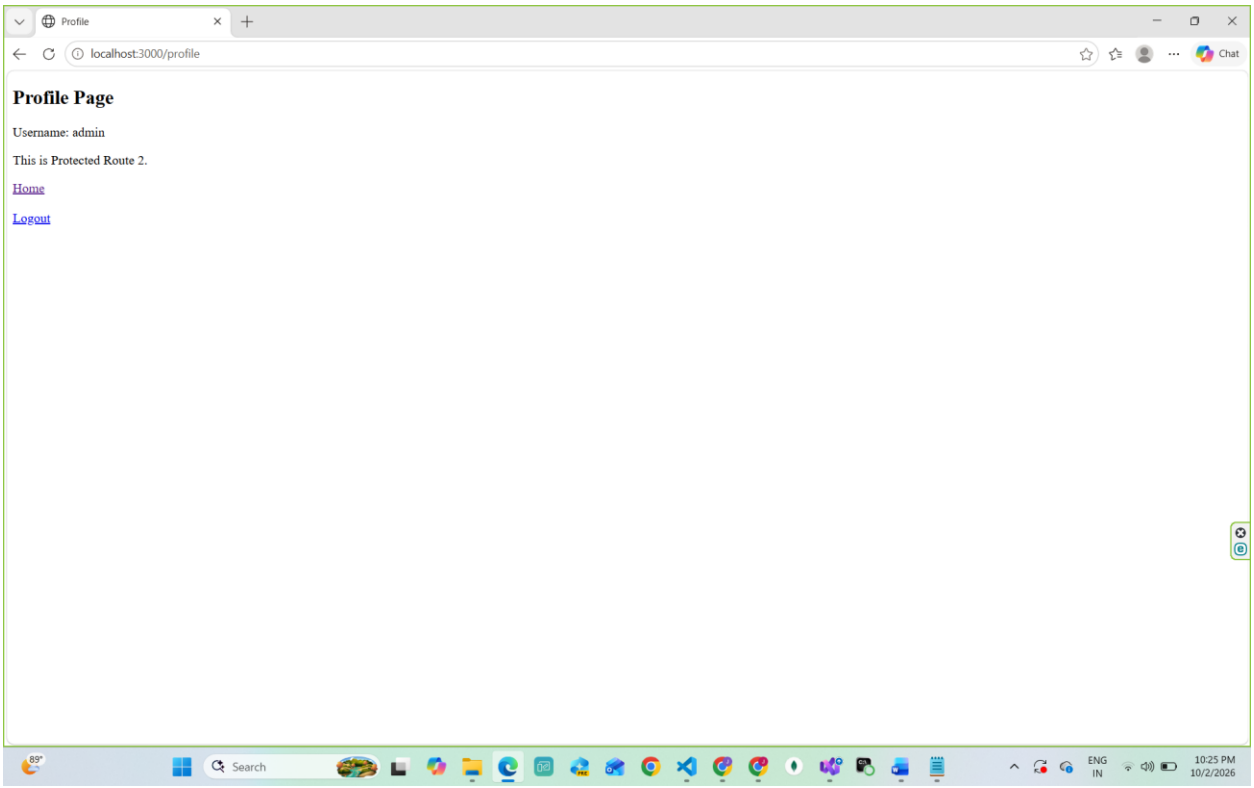

Registration Result
localhost:3000/register
Chat
Registration Details
Field
Value
Username
joy
Email
j@gmail.com
Gender
Male
Hobbies
Reading
Profile Picture
Download Profile Picture
Other Pictures
Download
Back to Registration
Downloads
1790959636880-Screenshot 2026-10-02 221651.png
Open file
1790959636852-Screenshot 2026-10-02 221651.png
Open file
Search
ENG
IN
10:19 PM
10/2/2026

2) Express Login application with file session store. Functionalities are login, 2 protected routes, Logout.

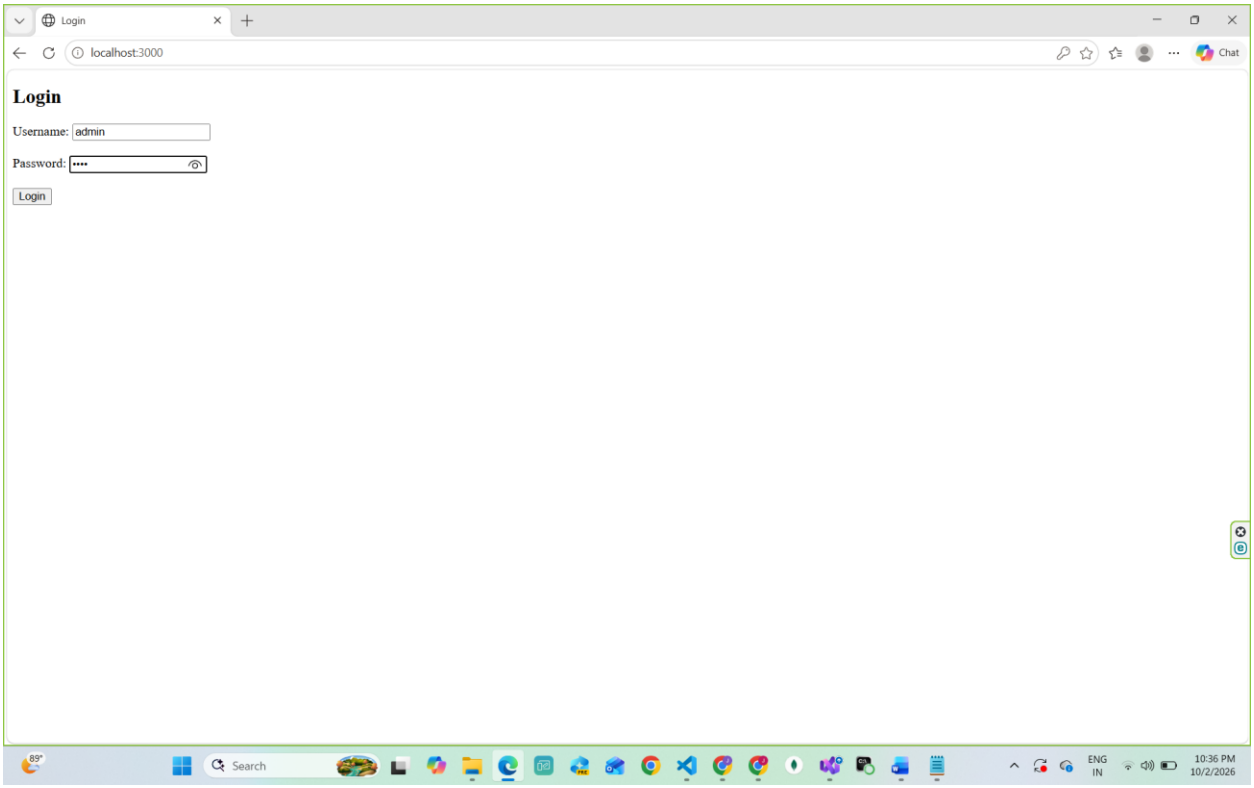


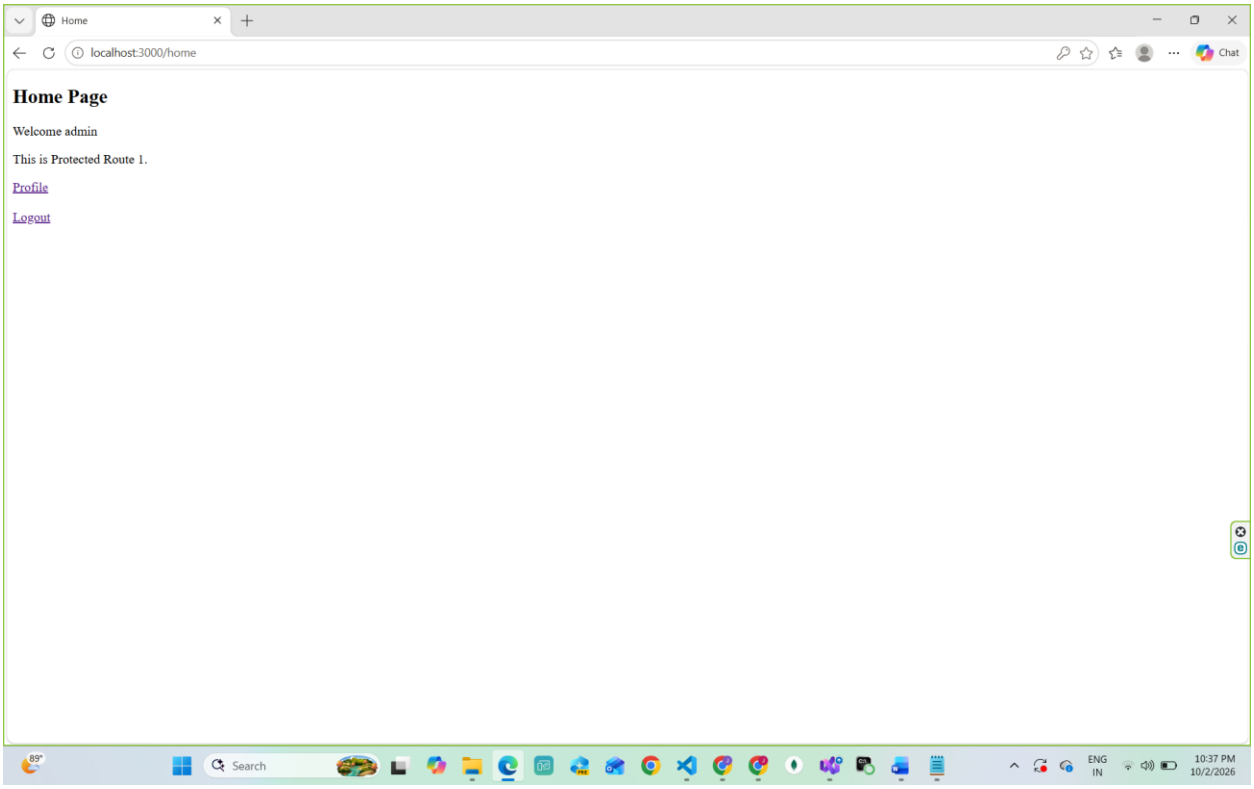


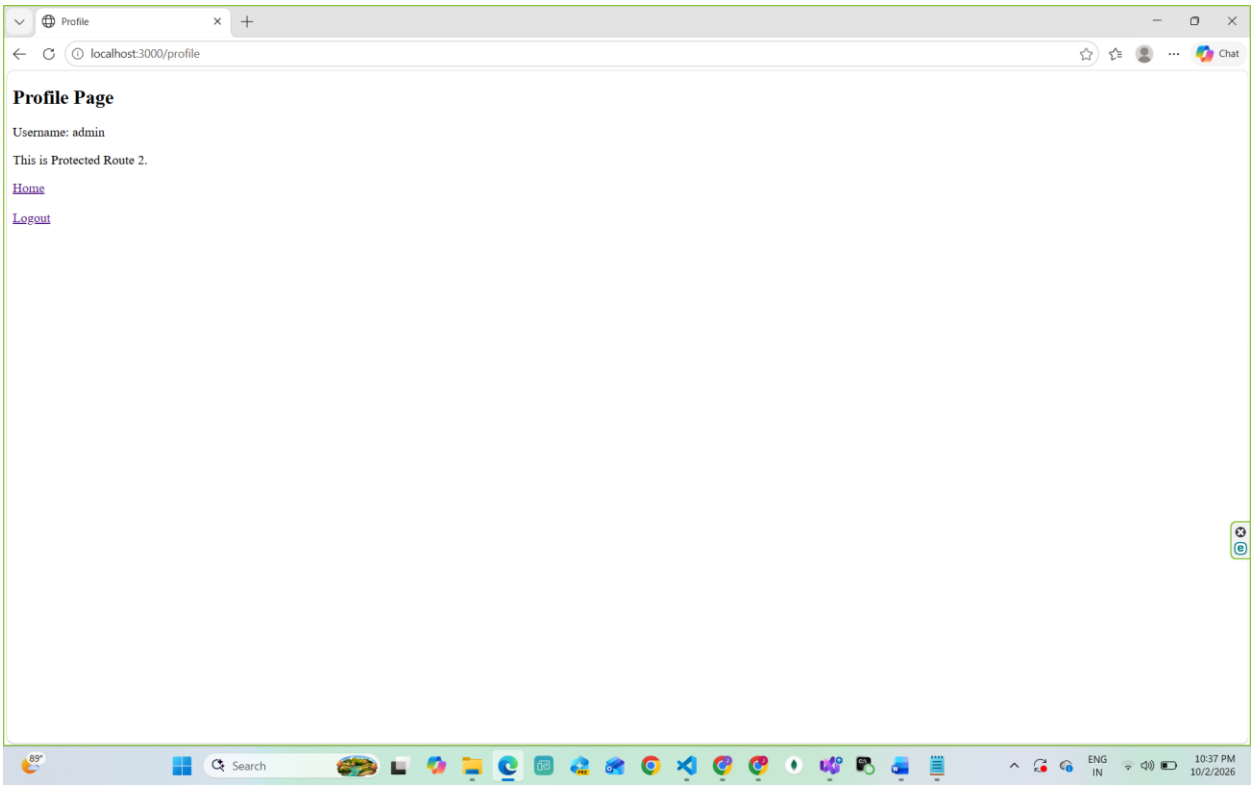

Home
localhost:3000/home
Chat
Home Page
Welcome admin
This is Protected Route 1.
Profile
Logout
Search
ENG
IN
10:25 PM
10/2/2026

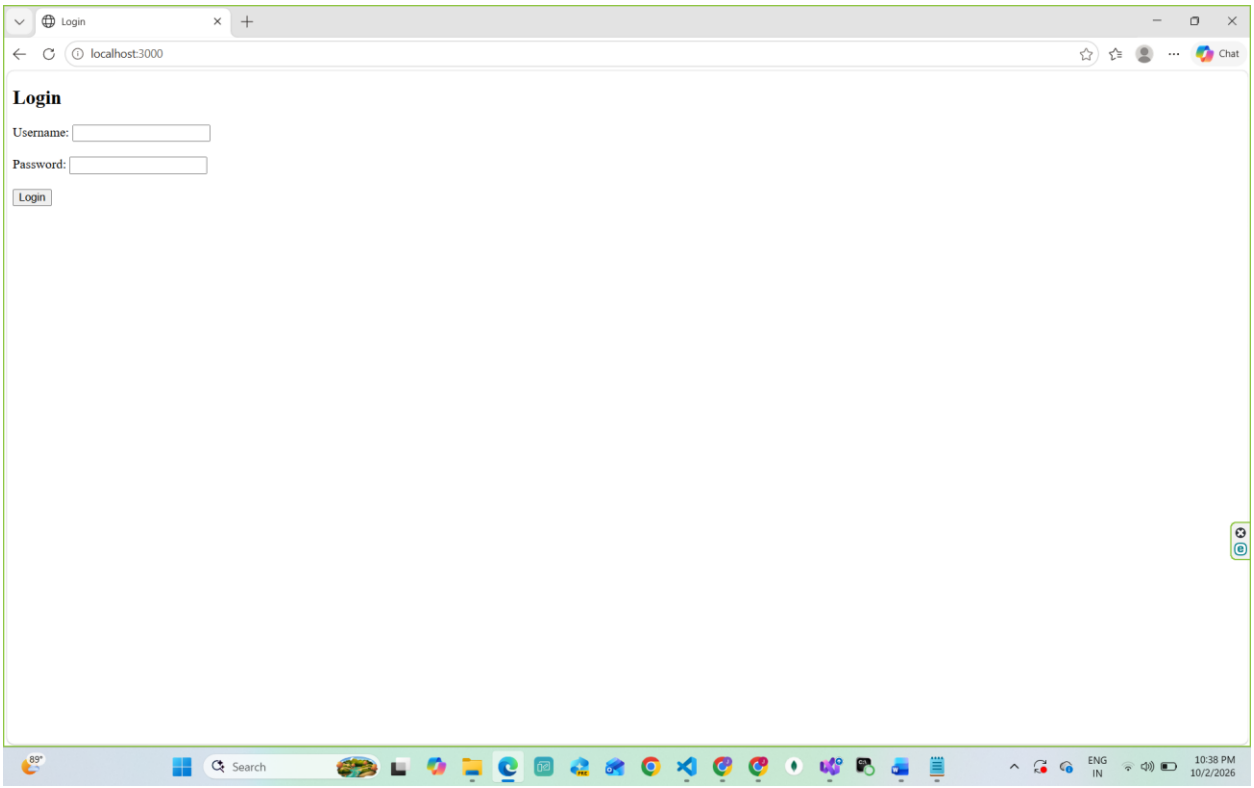

Profile
localhost:3000/profile
Chat
Profile Page
Username: admin
This is Protected Route 2.
Home
Logout
Search
ENG
IN
10:25 PM
10/2/2026

3) Express Login application with redis session store. Functionalities are login, 2 protected routes, Logout.

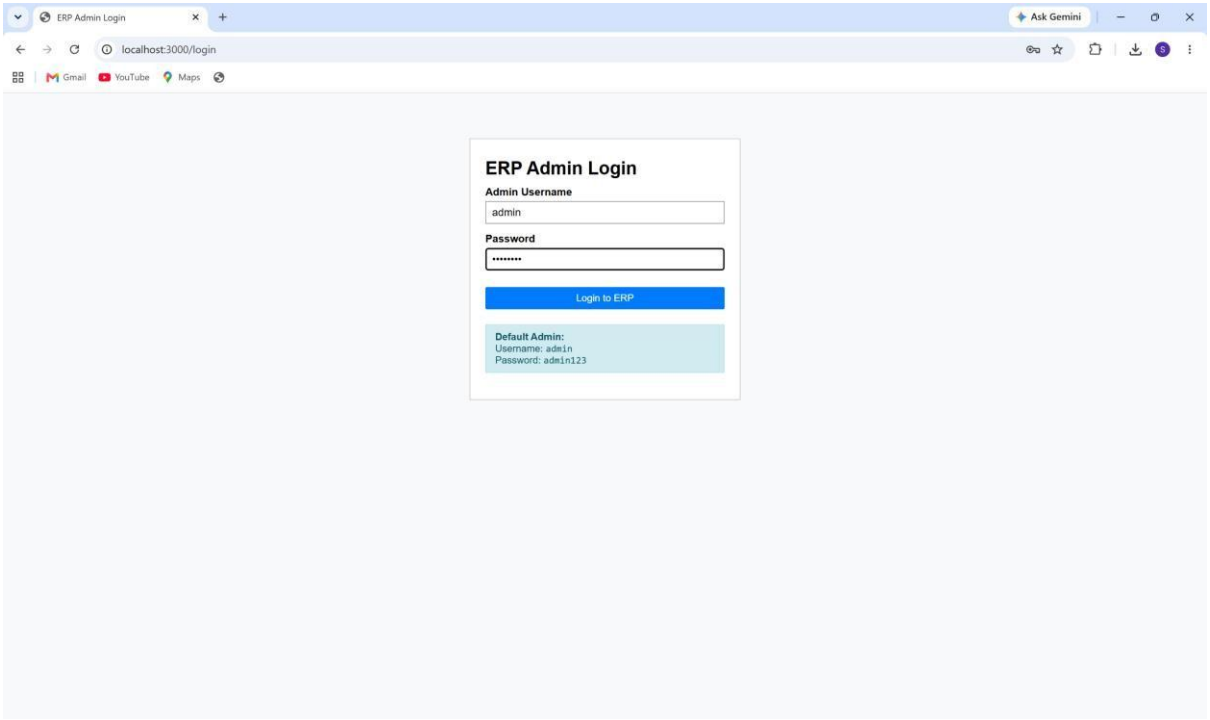


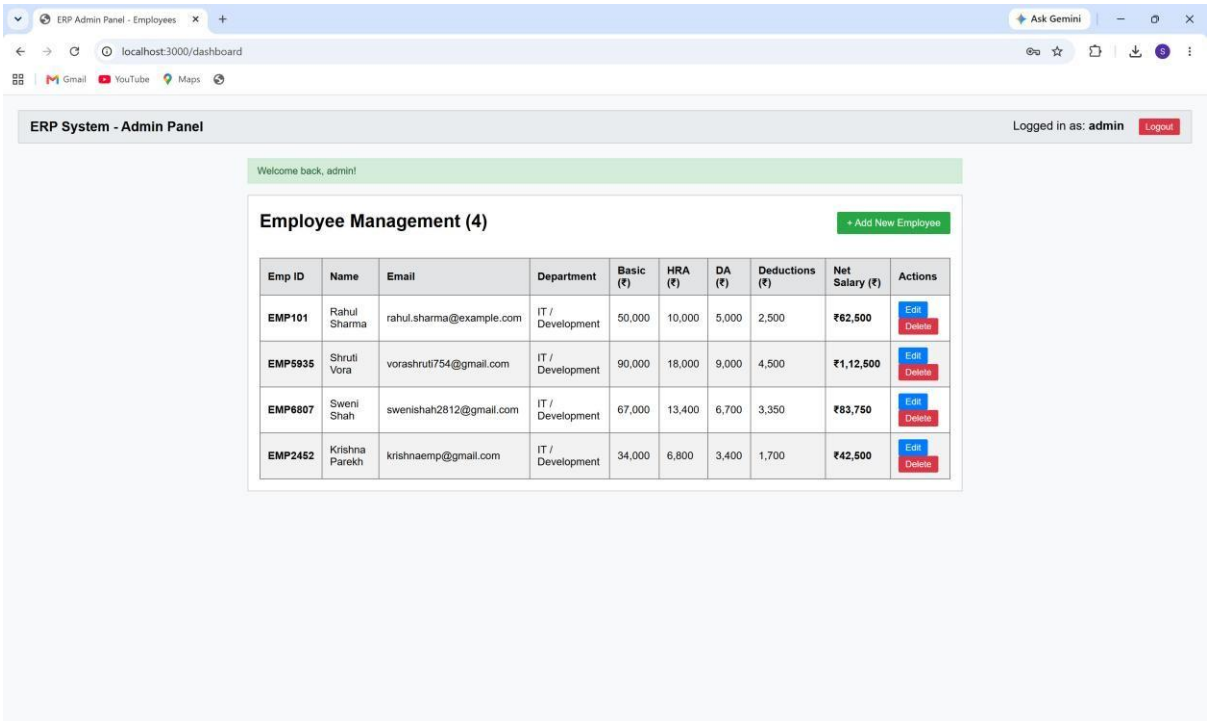


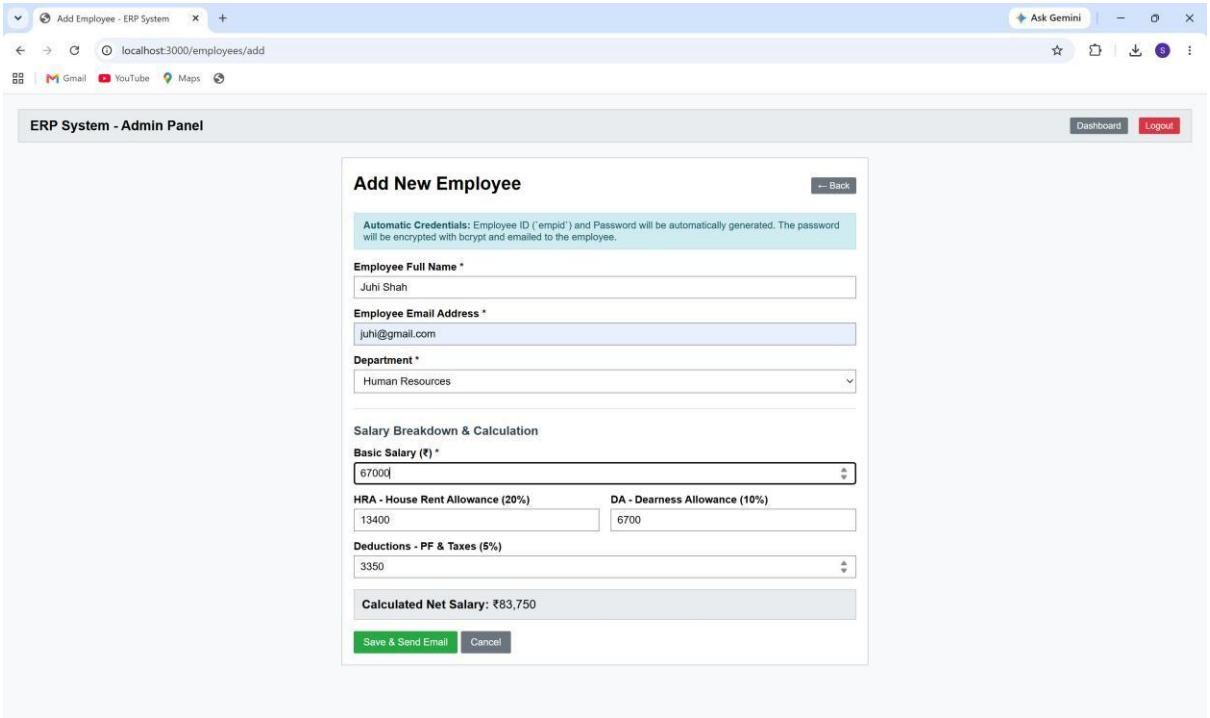

Profile
localhost:3000/profile
Chat
Profile Page
Username: admin
This is Protected Route 2.
Home
Logout
Search
ENG
IN
10:37 PM
10/2/2026

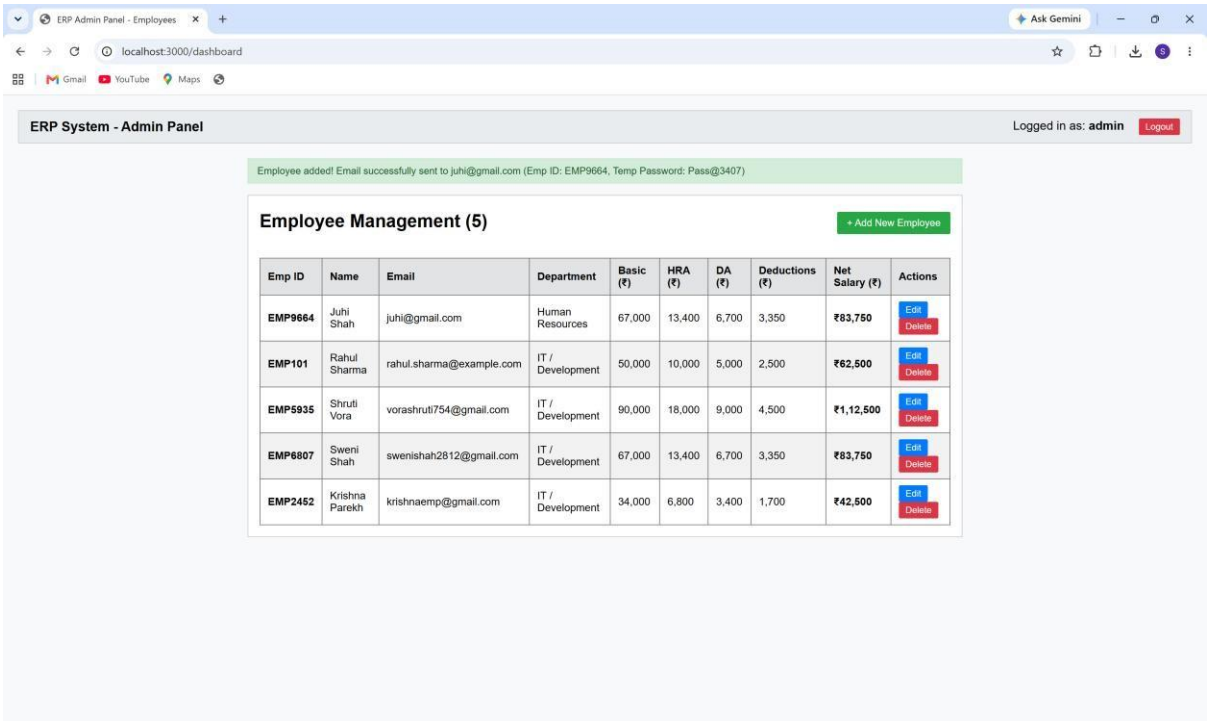

Login
localhost:3000
Chat
Login
Username:
Password:
Login
Search
ENG
IN
10:38 PM
10/2/2026

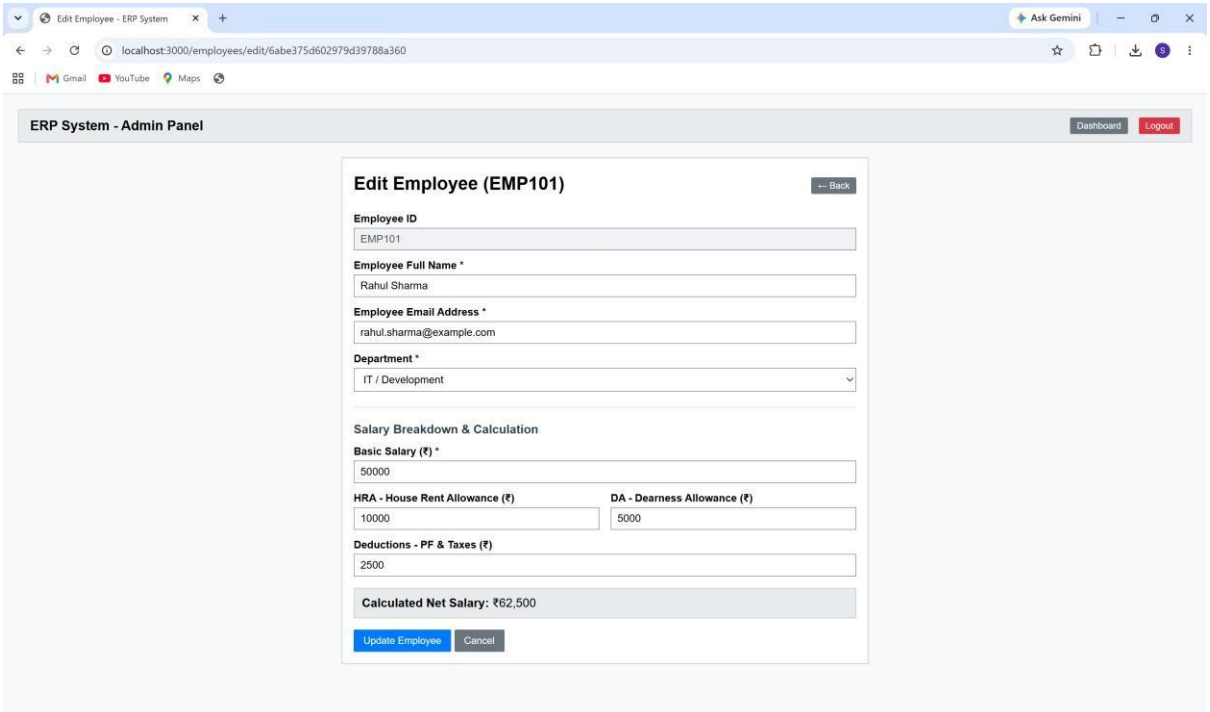
4) Develop login for an ADMIN panel of ERP system with session, CRUD operations for employees table (generate empid and password) with salary calculation with mongoose, express and EJS template engine. Send email to employee when inserted. Encrypt the password. Logout.

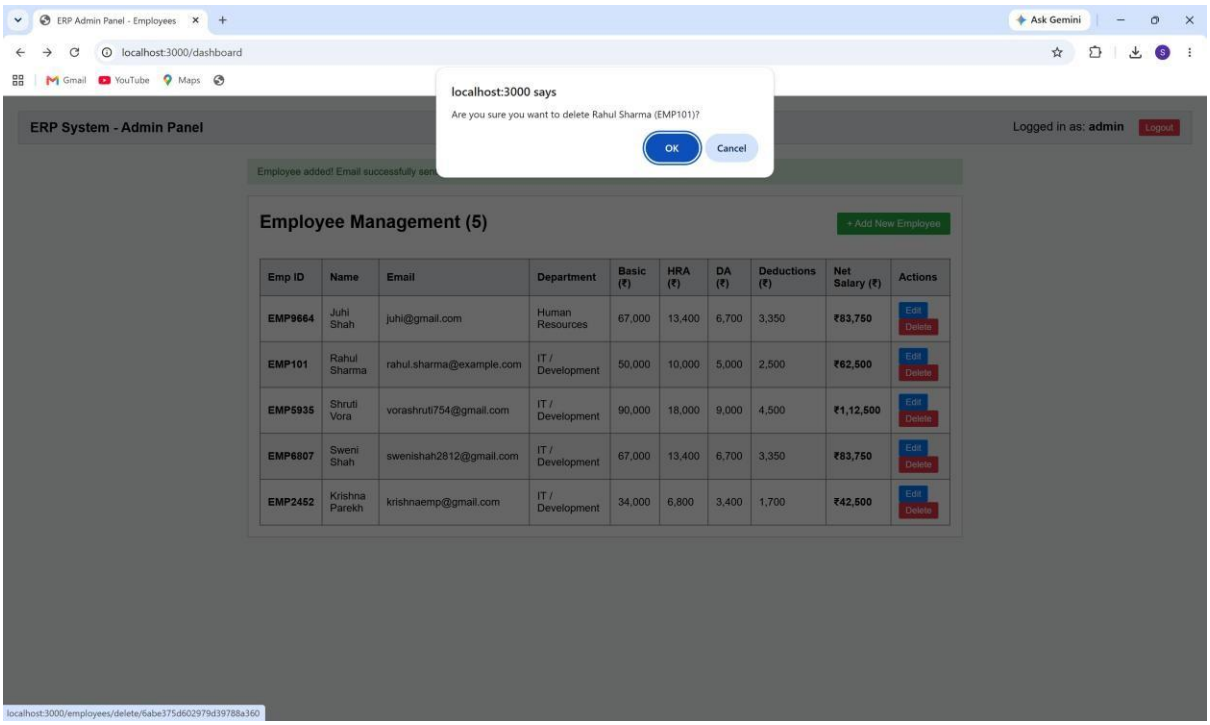



ERP Admin Panel - Employees
Ask Gemini
localhost:3000/dashboard
Gmail
YouTube
Maps
ERP System - Admin Panel
Logged in as: admin
Logout
Welcome back, admin!
Employee Management (4)
+ Add New Employee
Emp ID
Name
Email
Department
Basic (₹)
HRA (₹)
DA (₹)
Deductions (₹)
Net Salary (₹)
Actions
EMP101
Rahul Sharma
rahul.sharma@example.com
IT / Development
50,000
10,000
5,000
2,500
₹62,500
Edit
Delete
EMP5935
Shruti Vora
vorashruti754@gmail.com
IT / Development
90,000
18,000
9,000
4,500
₹1,12,500
Edit
Delete
EMP6807
Sweni Shah
swenishah2812@gmail.com
IT / Development
67,000
13,400
6,700
3,350
₹83,750
Edit
Delete
EMP2452
Krishna Parekh
krishnaemp@gmail.com
IT / Development
34,000
6,800
3,400
1,700
₹42,500
Edit
Delete


Add Employee - ERP System
Ask Gemini
localhost:3000/employees/add
Gmail
YouTube
Maps
ERP System - Admin Panel
Dashboard
Logout
Add New Employee
← Back
Automatic Credentials: Employee ID (`empid`) and Password will be automatically generated. The password will be encrypted with bcrypt and emailed to the employee.
Employee Full Name *
Juhi Shah
Employee Email Address *
juhi@gmail.com
Department *
Human Resources
Salary Breakdown & Calculation
Basic Salary (₹) *
67000
HRA - House Rent Allowance (20%)
13400
DA - Dearness Allowance (10%)
6700
Deductions - PF & Taxes (5%)
3350
Calculated Net Salary: ₹83,750
Save & Send Email
Cancel


ERP Admin Panel - Employees
Ask Gemini
localhost:3000/dashboard
Gmail
YouTube
Maps
ERP System - Admin Panel
Logged in as: admin
Logout
Employee added! Email successfully sent to juhi@gmail.com (Emp ID: EMP9664, Temp Password: Pass@3407)
Employee Management (5)
+ Add New Employee
Emp ID
Name
Email
Department
Basic (₹)
HRA (₹)
DA (₹)
Deductions (₹)
Net Salary (₹)
Actions
EMP9664
Juhi Shah
juhi@gmail.com
Human Resources
67,000
13,400
6,700
3,350
₹83,750
Edit
Delete
EMP101
Rahul Sharma
rahul.sharma@example.com
IT / Development
50,000
10,000
5,000
2,500
₹62,500
Edit
Delete
EMP5935
Shruti Vora
vorashruti754@gmail.com
IT / Development
90,000
18,000
9,000
4,500
₹1,12,500
Edit
Delete
EMP6807
Sweni Shah
swenishah2812@gmail.com
IT / Development
67,000
13,400
6,700
3,350
₹83,750
Edit
Delete
EMP2452
Krishna Parekh
krishnaemp@gmail.com
IT / Development
34,000
6,800
3,400
1,700
₹42,500
Edit
Delete


Edit Employee - ERP System
Ask Gemini
localhost:3000/employees/edit/6abe375d602979d39788a360
Gmail
YouTube
Maps
ERP System - Admin Panel
Dashboard
Logout
Edit Employee (EMP101)
← Back
Employee ID
EMP101
Employee Full Name *
Rahul Sharma
Employee Email Address *
rahul.sharma@example.com
Department *
IT / Development
Salary Breakdown & Calculation
Basic Salary (₹) *
50000
HRA - House Rent Allowance (₹)
10000
DA - Dearness Allowance (₹)
5000
Deductions - PF & Taxes (₹)
2500
Calculated Net Salary: ₹62,500
Update Employee
Cancel

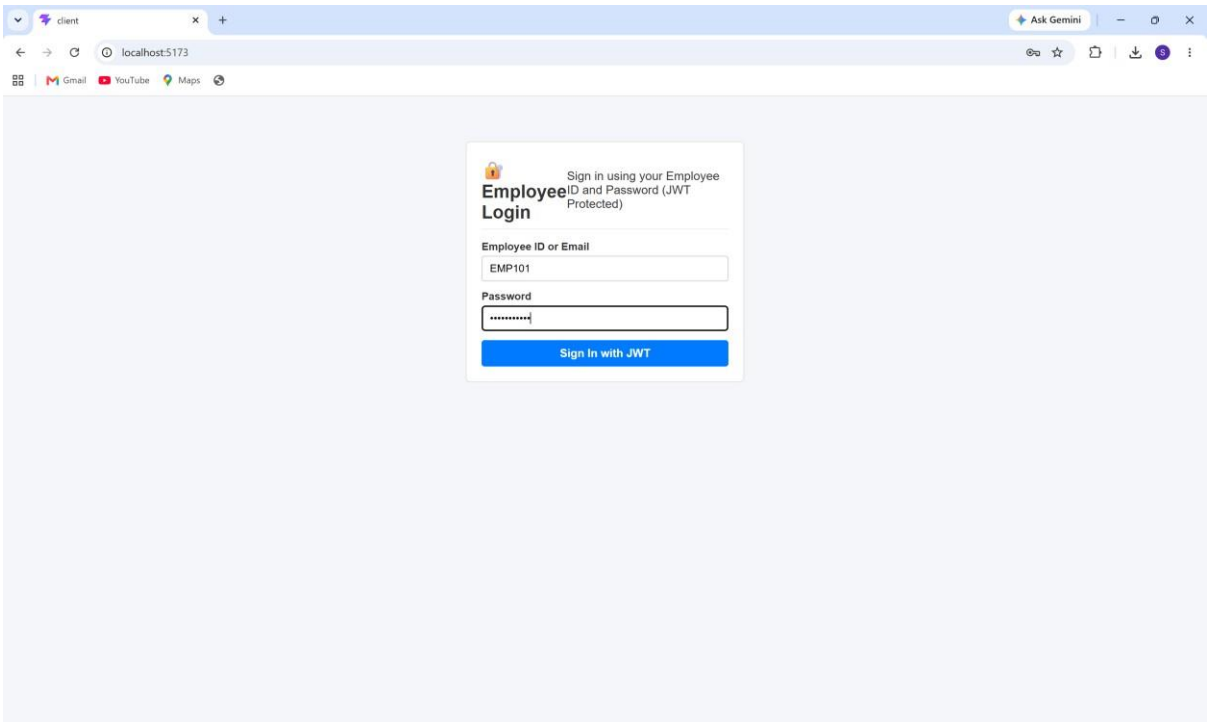

ERP Admin Panel - Employees
Ask Gemini
localhost:3000/dashboard
Gmail
YouTube
Maps
localhost:3000 says
Are you sure you want to delete Rahul Sharma (EMP101)?
OK
Cancel
ERP System - Admin Panel
Logged in as: admin
Logout
Employee added! Email successfully sen
Employee Management (5)
+ Add New Employee
Emp ID
Name
Email
Department
Basic (₹)
HRA (₹)
DA (₹)
Deductions (₹)
Net Salary (₹)
Actions
EMP9664
Juhi Shah
juhi@gmail.com
Human Resources
67,000
13,400
6,700
3,350
₹83,750
Edit
Delete
EMP101
Rahul Sharma
rahul.sharma@example.com
IT / Development
50,000
10,000
5,000
2,500
₹62,500
Edit
Delete
EMP5935
Shruti Vora
vorashruti754@gmail.com
IT / Development
90,000
18,000
9,000
4,500
₹1,12,500
Edit
Delete
EMP6807
Sweni Shah
swenishah2812@gmail.com
IT / Development
67,000
13,400
6,700
3,350
₹83,750
Edit
Delete
EMP2452
Krishna Parekh
krishnaemp@gmail.com
IT / Development
34,000
6,800
3,400
1,700
₹42,500
Edit
Delete
localhost:3000/employees/delete/6abe375d602979d39788a360

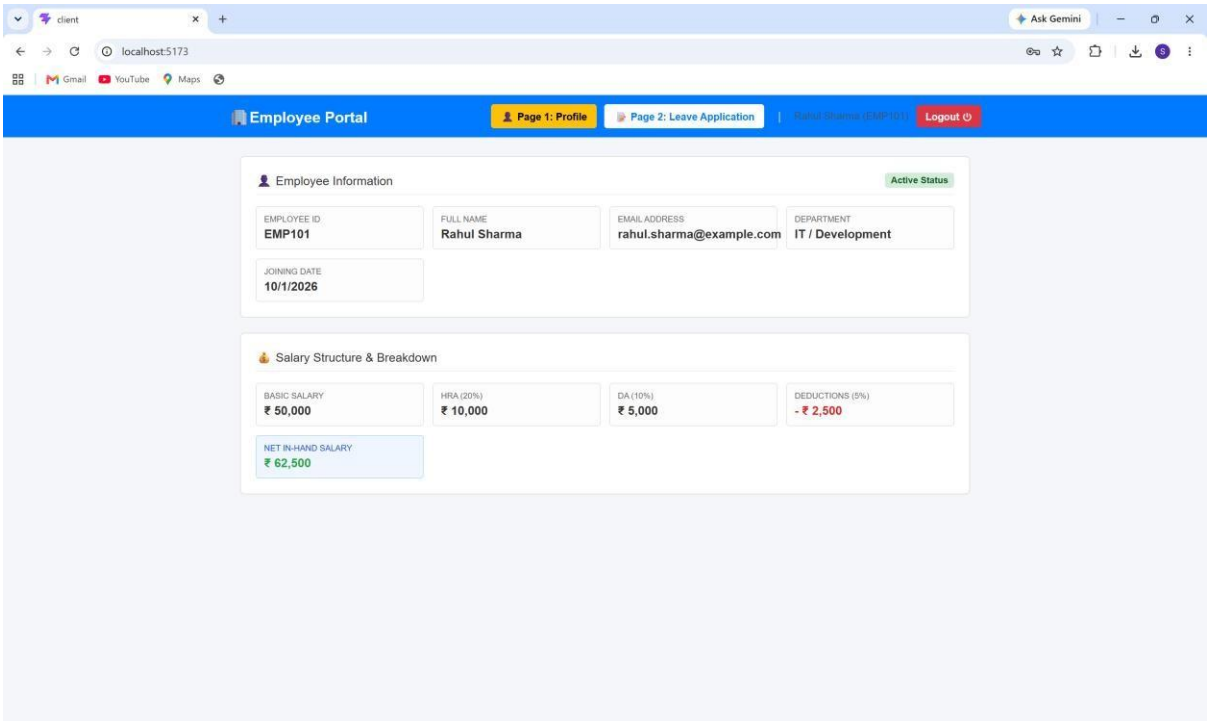
5) Develop employee site (refer Q4) login with JWT, for employees with mongoose, express and

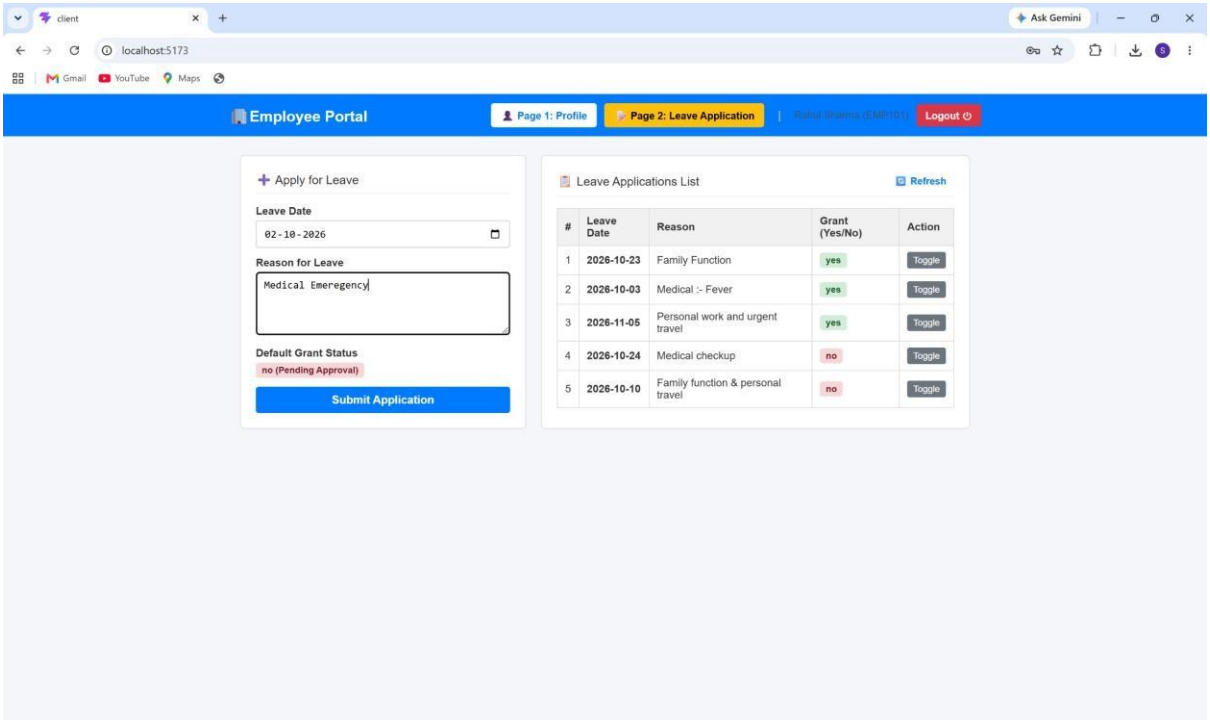
React.js. Create a home page with following links:

-Page1) display employee profile .

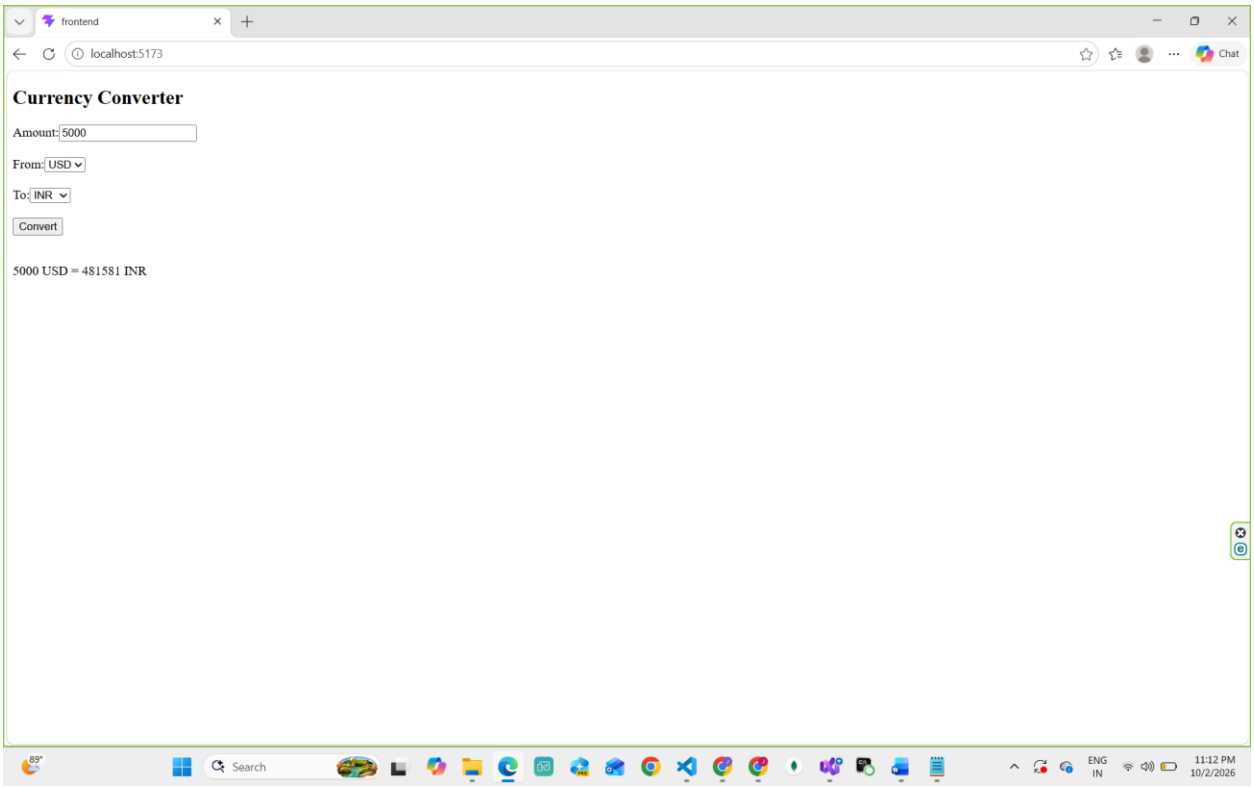
-Page2) - Application for leave (fields date, reason, grant yes/no) Add, List – Logout

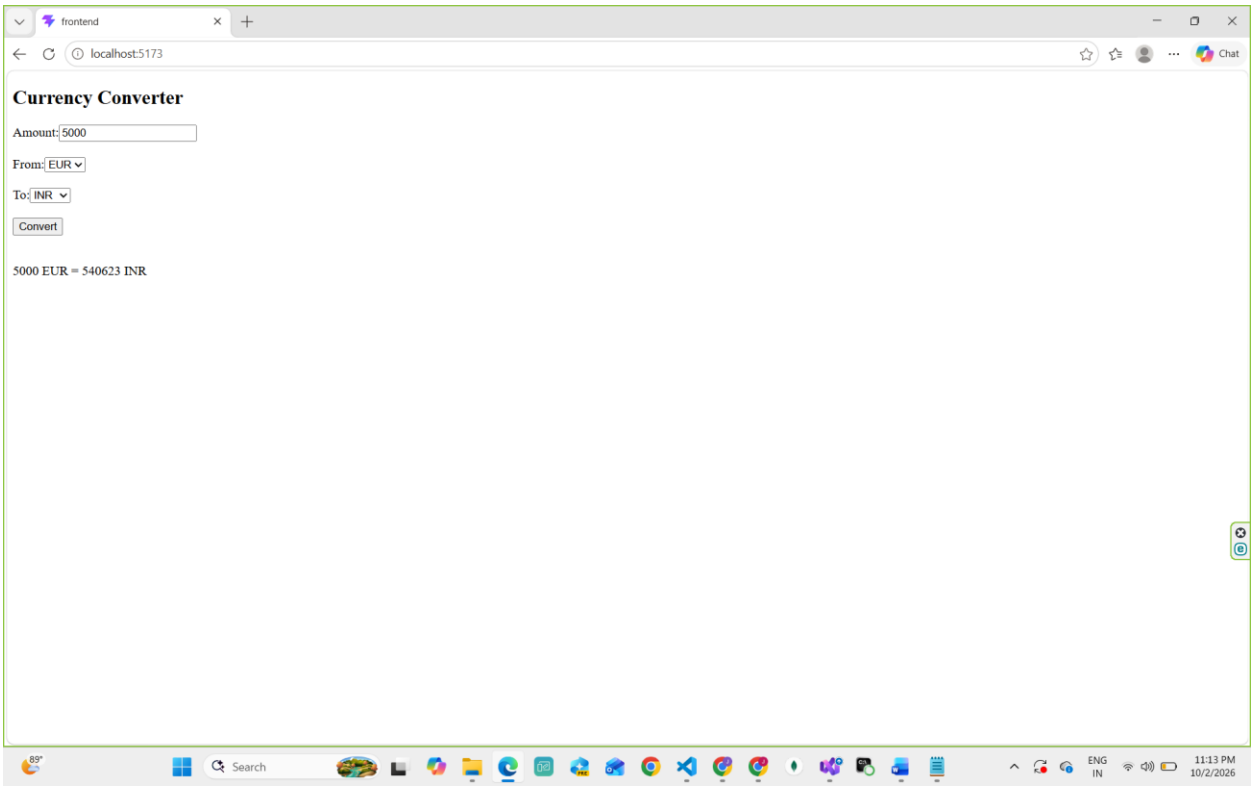



client
Ask Gemini
localhost:5173
Gmail
YouTube
Maps
Employee Portal
Page 1: Profile
Page 2: Leave Application
Logout
Employee Information
Active Status
EMPLOYEE ID
EMP101
FULL NAME
Rahul Sharma
EMAIL ADDRESS
rahul.sharma@example.com
DEPARTMENT
IT / Development
JOINING DATE
10/1/2026
Salary Structure & Breakdown
BASIC SALARY
₹ 50,000
HRA (20%)
₹ 10,000
DA (10%)
₹ 5,000
DEDUCTIONS (5%)
- ₹ 2,500
NET IN-HAND SALARY
₹ 62,500

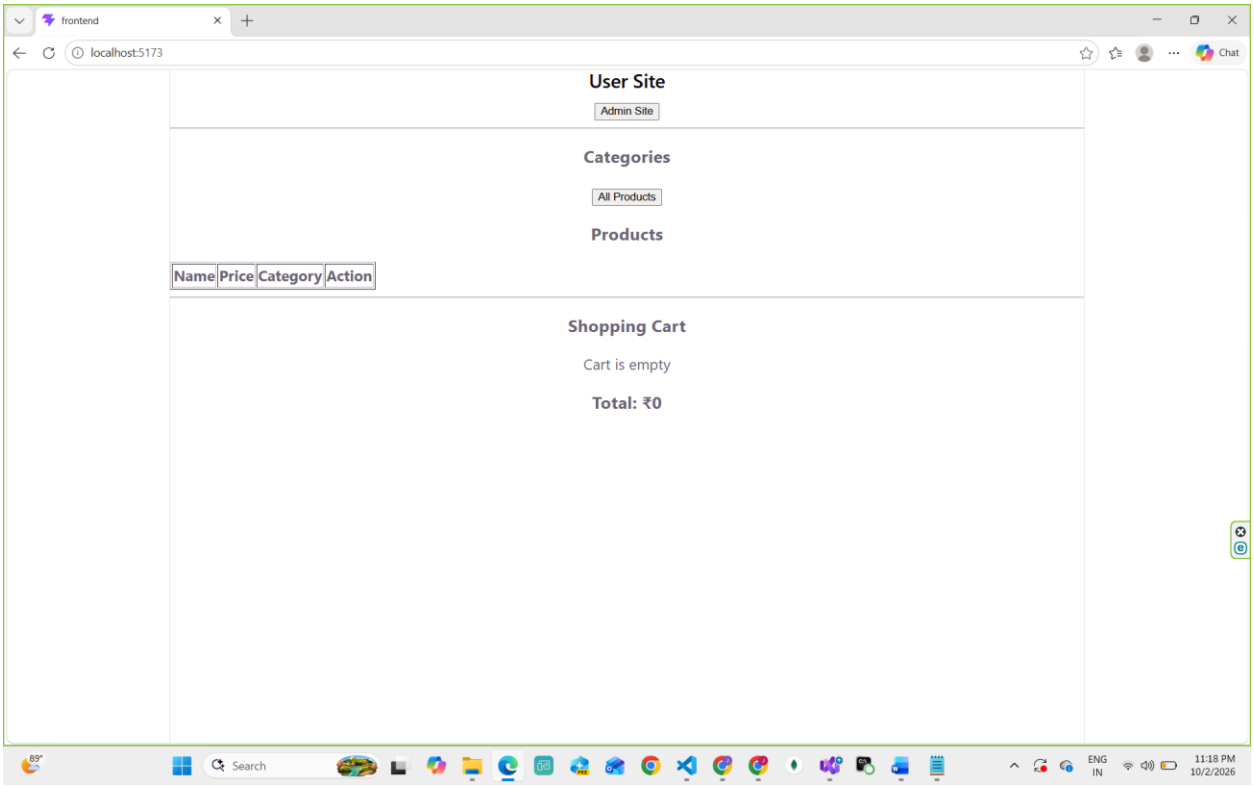

client
Ask Gemini
localhost:5173
Gmail
YouTube
Maps
Employee Portal
Page 1: Profile
Page 2: Leave Application
Logout
Apply for Leave
Leave Date
02-10-2026
Reason for Leave
Medical Emeregency
Default Grant Status
no (Pending Approval)
Submit Application
Leave Applications List
Refresh
Leave Date Reason Grant (Yes/No) Action
1 2026-10-23 Family Function yes Toggle
2 2026-10-03 Medical :- Fever yes Toggle
3 2026-11-05 Personal work and urgent travel yes Toggle
4 2026-10-24 Medical checkup no Toggle
5 2026-10-10 Family function & personal travel no Toggle

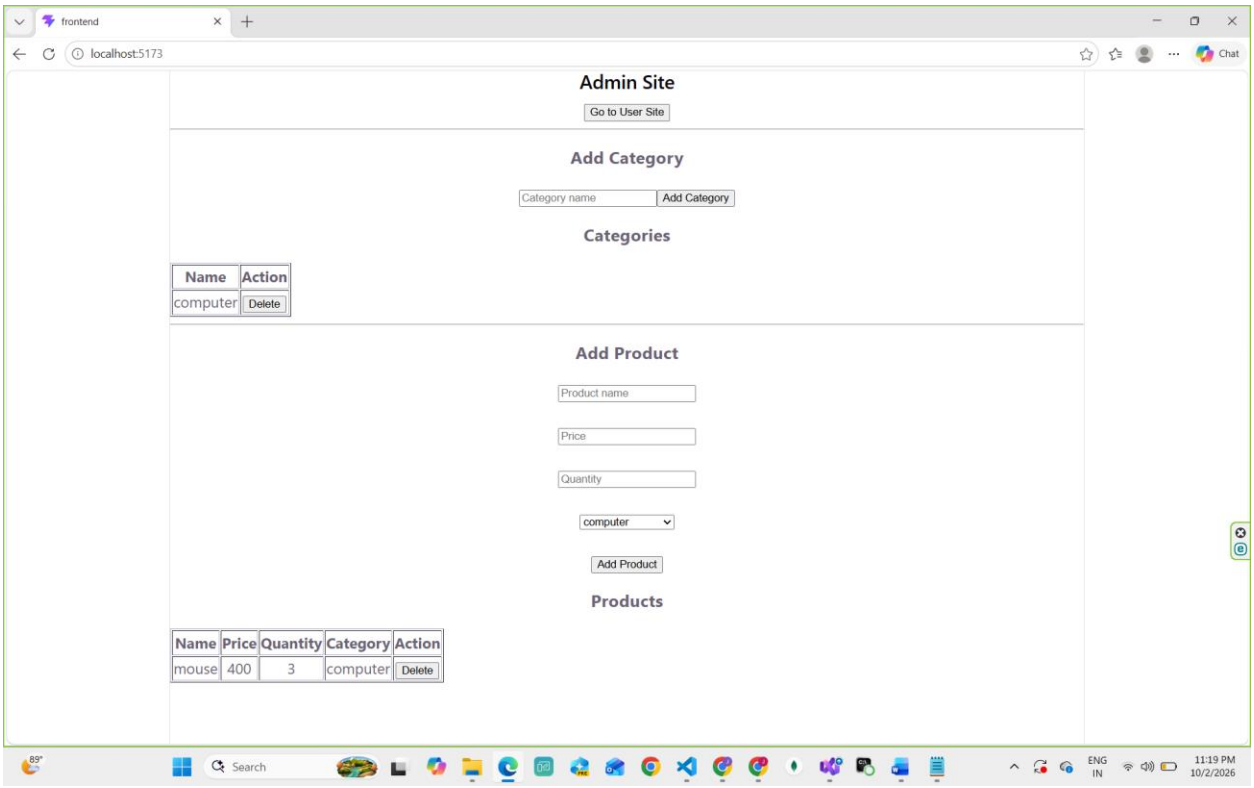
6) Call any free api for a useful utility from frontend and backend.

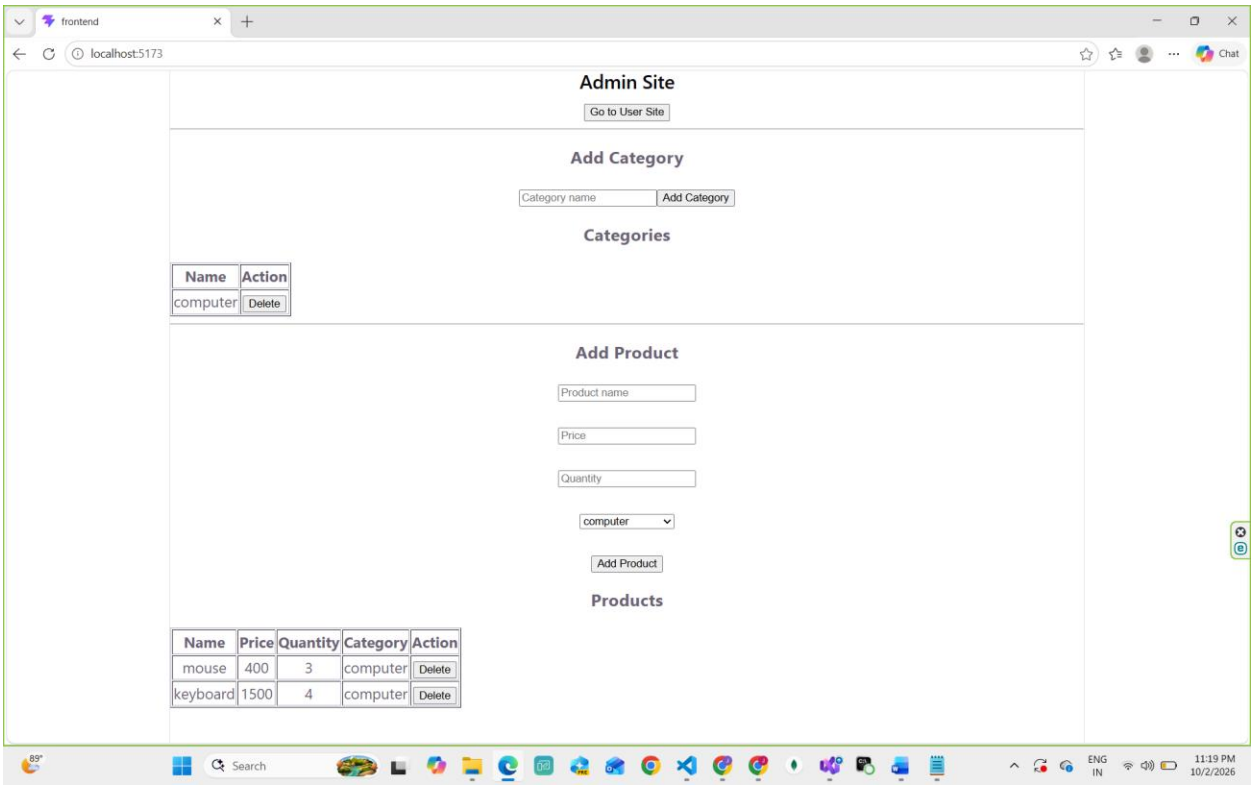


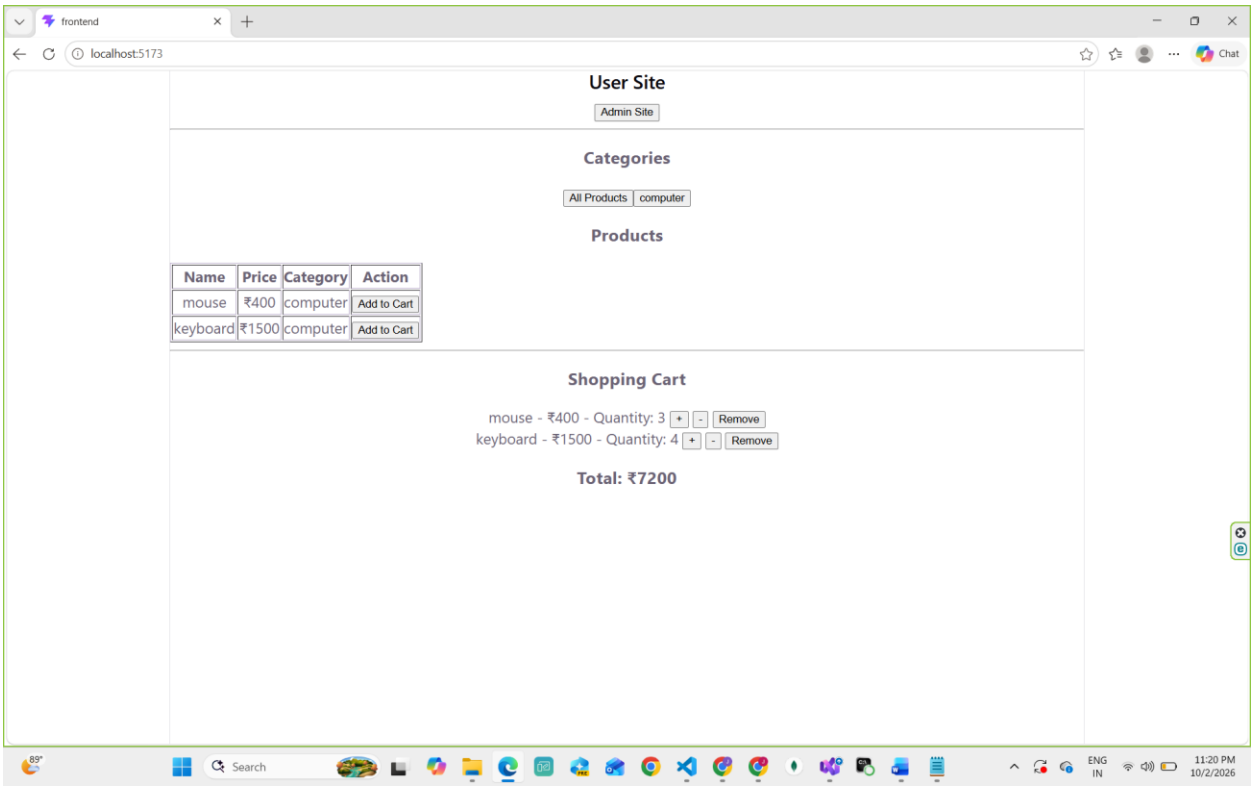


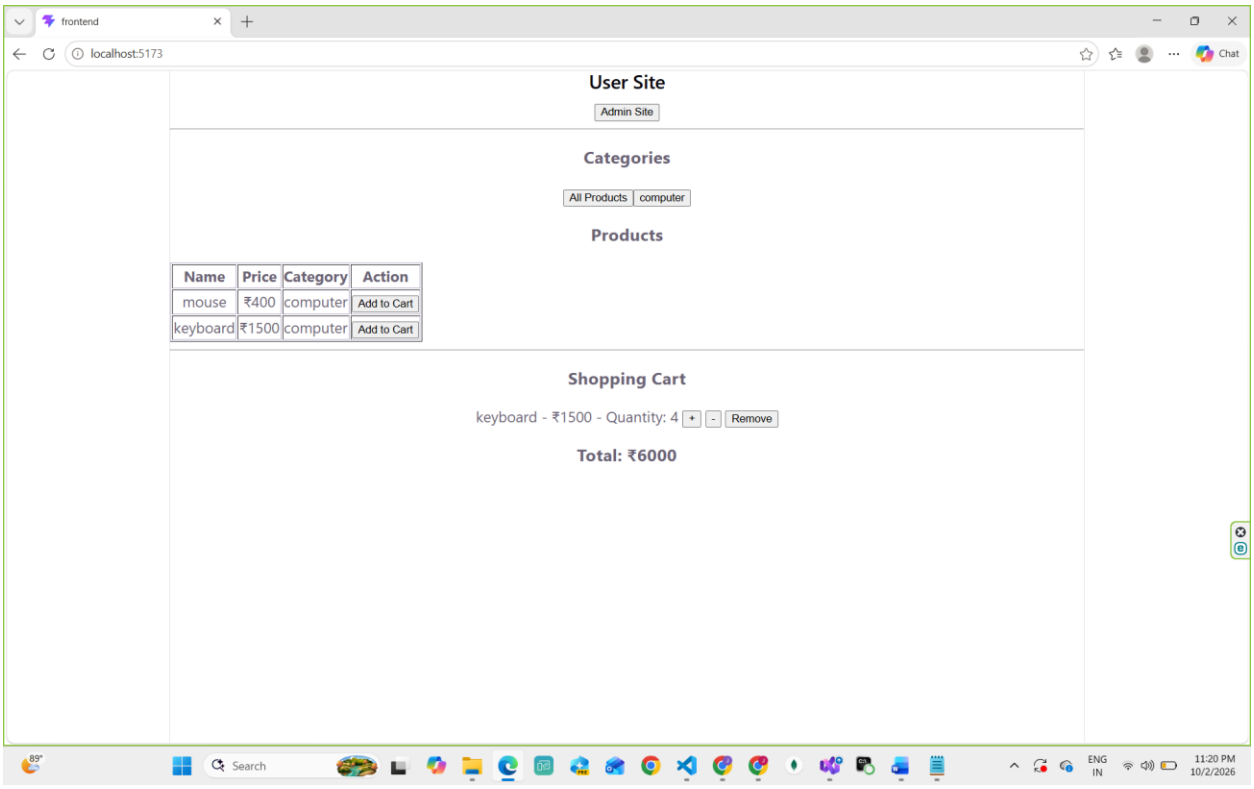
7) Shopping cart with admin site and user site with (2 level) Category and Products using MERN stack.






frontend
localhost:5173
Chat
Admin Site
Go to User Site
Add Category
Category name
Add Category
Categories
Name
Action
computer
Delete
Add Product
Product name
Price
Quantity
computer
Add Product
Products
Name
Price
Quantity
Category
Action
mouse
400
3
computer
Delete
keyboard
1500
4
computer
Delete
Search
ENG
IN
11:19 PM
10/2/2026


frontend
localhost:5173
Chat
User Site
Admin Site
Categories
All Products
computer
Products
Name
Price
Category
Action
mouse
₹400
computer
Add to Cart
keyboard
₹1500
computer
Add to Cart
Shopping Cart
mouse - ₹400 - Quantity: 3
+
-
Remove
keyboard - ₹1500 - Quantity: 4
+
-
Remove
Total: ₹7200
Search
ENG
IN
11:20 PM
10/2/2026


frontend
localhost:5173
Chat
User Site
Admin Site
Categories
All Products
computer
Products
Name
Price
Category
Action
mouse
₹400
computer
Add to Cart
keyboard
₹1500
computer
Add to Cart
Shopping Cart
keyboard - ₹1500 - Quantity: 4
+
-
Remove
Total: ₹6000
Search
ENG
IN
11:20 PM
10/2/2026

8) Crud operation for Student collection using Sequelize. (Express+ Sequelize + React)

